

Q1

Photosynthesis is the process by which green plants use light energy to make chemical energy in the form of glucose from carbon dioxide and water. The word equation is carbon dioxide + water → glucose + oxygen.

Q4

Deoxygenated blood enters the right atrium through the vena cava. It passes through the tricuspid valve into the right ventricle, then through the pulmonary valve into the pulmonary artery and lungs. Oxygenated blood returns through the pulmonary veins to the left atrium, passes through the bicuspid (mitral) valve into the left ventricle, and leaves through the aortic valve into the aorta.

Q2

The chloroplast is the organelle mainly involved in photosynthesis. Chlorophyll is the main pigment that absorbs light energy.

Q6

Food passes from the mouth to the oesophagus, stomach, small intestine, large intestine and finally the rectum and anus. The stomach churns food and begins protein digestion. The small intestine completes digestion and absorbs most nutrients through its villi. The liver produces bile, which helps emulsify fats, while the pancreas releases digestive enzymes.

Q6 continued

Most absorption occurs in the small intestine because its lining has many villi and microvilli, giving a very large surface area and a rich blood supply. Digested nutrients pass through the intestinal wall into blood or lymph for transport around the body.

Q11(b)

Plant B should be moved to a brighter position with suitable light. More light allows its chloroplasts to absorb more energy for photosynthesis and can help the plant develop stronger, greener leaves.

Q7

A nephron consists of Bowman's capsule containing the glomerulus, followed by the proximal convoluted tubule, loop of Henle, distal convoluted tubule and collecting duct. Filtration occurs at the glomerulus, while useful substances and water are selectively reabsorbed along the tubule.

Q3

Photosynthesis mainly occurs in chloroplasts. The light-dependent reactions capture light energy and produce energy-rich molecules. The light-independent reactions use this energy to help form glucose from carbon dioxide.

Q10

Xylem vessels are long, hollow tubes with no end walls between many cells, allowing water to move continuously. Their walls contain lignin, which strengthens the vessels and prevents collapse under tension.

Q12

Ventilation per minute = tidal volume $\times$ breathing rate. Therefore, $0.5 \text{ L} \times 12 = 6.0 \text{ L per minute}$.

Q13

Alveolar ventilation = (tidal volume – dead space) $\times$ breathing rate. Therefore, $(0.5 - 0.15) \times 12 = 0.35 \times 12 = 4.2 \text{ L per minute}$.

Q99

This is an extra answer that does not correspond to any question in the question paper. It is deliberately included to test unmatched-answer handling.